

一、判断题（每小题 2 分，共 20 分）

| 题号 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
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二、填空题（每空 2 分，共 20 分）

1. -2    2. E    3. 1    4. 线性无关    线性相关    有    5.  $(1, 2, 3, 4)^T + k(0, 1, 2, 3)^T$   
 6. 0, 0, 2     $k_1\alpha_1 + k_2\alpha_2 (\lambda = 0)$      $k_3\alpha_3 (\lambda = 2)$     ( $k_1, k_2$  不全为 0,  $k_3 \neq 0$ )

三、计算题（每小题 9 分，54 分）

1. 解：构造  $D' = \begin{vmatrix} 3 & 0 & 4 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & -7 & 0 & 0 \\ -1 & 1 & -1 & 1 \end{vmatrix} = 7 \begin{vmatrix} 3 & 4 & 0 \\ 2 & 2 & 2 \\ -1 & -1 & 1 \end{vmatrix} = 7 \begin{vmatrix} 3 & 4 & 0 \\ 4 & 4 & 2 \\ 0 & 0 & 1 \end{vmatrix} = 7 \begin{vmatrix} 3 & 4 \\ 4 & 4 \end{vmatrix} = -28$

或者  $M_{41} = -56$ ,  $M_{42} = 0$ ,  $M_{43} = 42$ ,  $M_{44} = -14$ , 故所求  $= -28$ . .....9'

2. 解：

$$(A|E) = \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 0 & 2 & -1 & 1 & -1 & 1 \\ 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[ \begin{array}{ccc|ccc} 0 & 0 & 3 & 1 & -1 & -1 \\ 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 0 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -\frac{1}{3} & \frac{1}{3} & \frac{2}{3} \end{array} \right] \rightarrow \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{array} \right]$$

.....8'

故  $A^{-1} = \frac{1}{3} \begin{bmatrix} 0 & 3 & -3 \\ 1 & -1 & 2 \\ -1 & 1 & 1 \end{bmatrix}$  .....9'

$$3. \text{ 解: } (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & 3 & \alpha+2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & \alpha+1 \end{bmatrix} \dots\dots\dots 6'$$

故当  $\alpha+1=0$ ，即  $\alpha=-1$  时， $r(\alpha_1, \alpha_2, \alpha_3)=2$ ， $\alpha_1, \alpha_2, \alpha_3$  线性相关；

当  $\alpha+1 \neq 0$ ，即  $\alpha \neq -1$  时， $r(\alpha_1, \alpha_2, \alpha_3)=3$ ， $\alpha_1, \alpha_2, \alpha_3$  线性无关………9'

4. 解:

$$\bar{A} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & 1 & -3 & 0 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 5 & 4 & 3 & 3 & -1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & -1 & -5 & -2 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \dots\dots\dots 5'$$

$$\text{令 } x_3 = x_4 = x_5 = 0, \text{ 可得特解 } \eta_0 = (-2, 3, 0, 0, 0)^T \dots\dots\dots 6'$$

分 别 令  $\begin{pmatrix} x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ ，可得导出组的基础解系：

$$\xi_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \xi_3 = \begin{pmatrix} 5 \\ -6 \\ 0 \\ 0 \\ 1 \end{pmatrix} \dots\dots\dots 8'$$

$$\text{故通解为: } \eta = \begin{pmatrix} -2 \\ 3 \\ 0 \\ 0 \\ 0 \end{pmatrix} + k_1 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + k_3 \begin{pmatrix} 5 \\ -6 \\ 0 \\ 0 \\ 1 \end{pmatrix} \dots\dots\dots 9'$$

5. 解:

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = \begin{bmatrix} 3 & 1 & 1 & 4 & 2 \\ 1 & 1 & -1 & 2 & 0 \\ 2 & 1 & 0 & 3 & 1 \\ 5 & 2 & 1 & 7 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & -1 & 2 & 0 \\ 0 & -1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & -2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \dots\dots\dots 6'$$

$r(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = 2$  向量组  $\alpha_1, \alpha_2$  是一个极大线性无关组;

$$\alpha_3 = \alpha_1 - 2\alpha_2, \alpha_4 = \alpha_1 + \alpha_2, \alpha_5 = \alpha_1 - \alpha_2, \dots\dots\dots 9'$$

6. 解  $|\lambda E - A| = \begin{vmatrix} \lambda - 1 & 0 & 0 \\ -1 & \lambda - 2 & -2 \\ -1 & -1 & \lambda - 3 \end{vmatrix} = (\lambda - 1)^2(\lambda - 4)$

故特征值为 1, 1, 4

对  $\lambda_1 = \lambda_2 = 1$ , 由  $(E - A)x = O$ ,  $E - A = \begin{bmatrix} 0 & 0 & 0 \\ -1 & -1 & -2 \\ -1 & -1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$  求得基础解系为

$$\xi_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \xi_2 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}, \text{ 故 } A \text{ 可以相似对角化} \dots\dots\dots 6'$$

$\lambda_3 = 4$  求得基础解系为:  $\xi_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix};$

故可逆矩阵为:  $P = \begin{bmatrix} -1 & -2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

使得  $P^{-1}AP = \Lambda$ .  $\dots\dots\dots 9'$

#### 四、证明题 (共 6 分)

证:  $(A^*)^T = (A^T)^* = (-A)^* = (-1)^{n-1} A^* = -A^* \dots\dots\dots 6'$

或者

$$(A^*)^T = (|A| A^{-1})^T = |A| (A^T)^{-1} = |A| (-A)^{-1} = -|A| A^{-1} = -A^* \dots\dots\dots 6'$$